

Target Translations Obstruct Gradient-Only Control of the Topological Singular-Set Operator

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Status. Candidate full counterexample, likely valid, to the question following Theorem 3.1 in the source paper.

1 Source question

Canevari and Orlandi construct

$$\mathbf{S} : (L^\infty \cap W^{1,k-1})(\Omega, \mathbb{R}^m) \longrightarrow L^1(\mathbb{R}^m; \mathbb{F}_{d-k}(\Omega; \pi_{k-1}(\mathcal{N}))) =: Y.$$

After proving a continuity estimate involving $\|u_1 - u_0\|_{L^k}$, they ask whether $\|\mathbf{S}(u_1) - \mathbf{S}(u_0)\|_Y$ can instead be bounded in terms of Sobolev norms of $\nabla u_1 - \nabla u_0$ [1].

with [2] Theorem 3.8]). Finally, (P₅) is a continuity estimate in the spirit of [20] Theorem 1], Statement (i). By applying the Hölder inequality to (P₅), for any $\Lambda > 0$ and any vector-valued maps $u_0, u_1 \in (L^\infty \cap W^{1,k})(\Omega, \mathbb{R}^m)$ satisfying $\|u_0\|_{L^\infty(\Omega)} \leq \Lambda$ and $\|u_1\|_{L^\infty(\Omega)} \leq \Lambda$ we obtain

$$(3.1) \quad \int_{\mathbb{R}^k} \mathbb{F}_\Omega(\mathbf{S}_y(u_1) - \mathbf{S}_y(u_0)) \, dy \leq C_\Lambda \|u_0 - u_1\|_{L^k(\Omega)} \left(\|\nabla u_0\|_{L^k(\Omega)}^{k-1} + \|\nabla u_1\|_{L^k(\Omega)}^{k-1} \right).$$

A natural question is whether the distance $\|\mathbf{S}(u_1) - \mathbf{S}(u_0)\|_Y$ can be bounded in terms of the Sobolev norms of $\nabla u_1 - \nabla u_0$. Notice that such an estimate is available for the Jacobian (see [20] Theorem 1], Statement (ii)). However, the answer is not known in general.

As is standard for a continuity estimate, “controlled by the gradient difference” means that the proposed right-hand side vanishes when all those norms vanish. The example below rules out every such estimate, including ones whose coefficient also depends on fixed L^∞ bounds or on individual derivative norms.

2 Proof intuition

The derivative forgets additive constants in the target, but $\mathbf{S}$ does not. In the sphere case, $\mathbf{S}_y(u)$ records the signed preimages of the point y . Replacing u by $u + c$ therefore translates the whole family in the parameter y , although it leaves the derivative unchanged. Choosing c large enough makes the two ranges disjoint. A positive-volume family of unit point chains remains in their difference, so its Y -norm cannot vanish.

3 Counterexample theorem

Theorem 1. *For every integer $k \geq 2$, let*

$$d = m = k, \quad \Omega = B_1(0) \subset \mathbb{R}^k, \quad \mathcal{N} = S^{k-1} \subset \mathbb{R}^k.$$

Put $c = 3e_1$ and define smooth maps

$$u_0(x) = x, \quad u_1(x) = x + c.$$

Then

$$\nabla u_1 - \nabla u_0 = 0$$

pointwise, but

$$\|\mathbf{S}(u_1) - \mathbf{S}(u_0)\|_Y \geq \frac{1}{4} |B_{1/2}(0)| > 0.$$

Consequently the source question has a negative answer in general: no estimate controlled only by Sobolev norms of $\nabla u_1 - \nabla u_0$, and vanishing when that difference is zero, can hold for $\mathbf{S}$.

Proof. Both maps belong to $L^\infty \cap W^{1,k-1}$ (indeed, they are affine), and their derivatives are both the identity matrix.

For $\mathcal{N} = S^{k-1} \subset \mathbb{R}^k$, the source chooses the exceptional complex $\mathcal{X} = \{0\}$ and the radial retraction $z \mapsto z/|z|$. Its smooth-map construction says that $\mathbf{S}_y(u)$ is the signed zero-dimensional preimage chain of y : each regular preimage carries its local degree in $\pi_{k-1}(S^{k-1}) \simeq \mathbb{Z}$ [1, Sections 1.2 and 3.2]. Since the two affine derivatives preserve orientation, there is one fixed $\varepsilon \in \{1, -1\}$, depending only on the source's generator convention, such that for almost every y

$$\mathbf{S}_y(u_0) = \begin{cases} \varepsilon \llbracket y \rrbracket, & y \in B_1(0), \\ 0, & y \notin B_1(0), \end{cases} \quad \mathbf{S}_y(u_1) = \begin{cases} \varepsilon \llbracket y - c \rrbracket, & y \in c + B_1(0), \\ 0, & y \notin c + B_1(0). \end{cases}$$

The possible sign will play no role.

Because $|c| = 3 > 2$, the two range balls are disjoint. In particular, for almost every $y \in B_{1/2}(0)$,

$$\mathbf{S}_y(u_1) - \mathbf{S}_y(u_0) = -\varepsilon \llbracket y \rrbracket \quad \text{in } \mathbb{F}_0(\Omega; \mathbb{Z}).$$

It remains only to keep the relative flat norm from losing this point through the boundary of Ω .

Define the compactly supported Lipschitz function

$$\varphi(x) = \min \left\{ \frac{1}{4}, \left(\frac{3}{4} - |x| \right)_+ \right\}.$$

Then $0 \leq \varphi \leq 1$, $\text{Lip}(\varphi) \leq 1$, $\text{spt } \varphi \Subset B_1$, and $\varphi = 1/4$ on $B_{1/2}$. By the source's characterization of the relative flat norm, any admissible decomposition of $\llbracket y \rrbracket$ has the form

$$\llbracket y \rrbracket = \partial P + Q \quad \text{modulo a chain supported in } \mathbb{R}^k \setminus \Omega,$$

and is charged by $\mathbf{M}(P \llcorner \Omega) + \mathbf{M}(Q \llcorner \Omega)$. Testing against φ (or smooth compactly supported approximations) gives, for $y \in B_{1/2}$,

$$\begin{aligned} \frac{1}{4} &= \llbracket y \rrbracket(\varphi) = P(d\varphi) + Q(\varphi) \\ &\leq \mathbf{M}(P \llcorner \Omega) + \mathbf{M}(Q \llcorner \Omega). \end{aligned}$$

Taking the infimum yields $\mathbb{F}_\Omega(\llbracket y \rrbracket) \geq 1/4$. Therefore

$$\|\mathbf{S}(u_1) - \mathbf{S}(u_0)\|_Y = \int_{\mathbb{R}^k} \mathbb{F}_\Omega(\mathbf{S}_y(u_1) - \mathbf{S}_y(u_0)) \, dy \geq \int_{B_{1/2}} \frac{1}{4} \, dy = \frac{1}{4} |B_{1/2}| > 0.$$

This contradicts every proposed gradient-difference estimate whose right-hand side vanishes at zero. $\square$

4 Scope, repair, and verification

The counterexample addresses the literal unrestricted question. It does not conflict with the source’s proved estimate, whose right-hand side contains $\|u_1 - u_0\|_{L^k}$. Nor does it rule out a gradient-based estimate after fixing target-translation freedom, for example by prescribing a common mean, a common trace, or one point value. Another natural repair is to compare the y -families only after compensating for the target translation.

The signed-preimage calculation is direct from the source’s smooth construction. The relative-norm argument uses exactly its quotient-norm characterization. The accompanying verification script checks the affine and disjoint-range identities in several dimensions; the proof above is valid for every $k \geq 2$. A separate audit is recorded in `VERIFICATION.md`.

5 Bounded novelty search

A search on 2026-08-17 used the exact question sentence, the paper title and authors, “topological singular set gradient difference”, and the companion-paper and survey literature. It found the source, Part II [2], the authors’ 2021 survey [3], and later applications of the same singular-chain operator, but no work explicitly answering this question or recording the target-translation obstruction. This is a bounded search, not a priority claim. Novelty confidence is moderate; mathematical-validity confidence is high.

References

- [1] G. Canevari and G. Orlandi, *Topological singular set of vector-valued maps, I: Applications to manifold-constrained Sobolev and BV spaces*, arXiv:1712.10203v2 (2018), especially Theorem 3.1 and Sections 1.2, 2.1, 3.2.
- [2] G. Canevari and G. Orlandi, *Topological singular set of vector-valued maps, II: Γ -convergence for Ginzburg–Landau type functionals*, Arch. Ration. Mech. Anal. 241 (2021), 117–193; arXiv:2003.01354.
- [3] G. Canevari and G. Orlandi, *Topological singularities for vector-valued Sobolev maps and applications*, Ann. Fac. Sci. Toulouse Math. 30 (2021), 327–351.